

SMART INDIA HACKATHON (SIH) • PROBLEM STATEMENT: SIH26068

Weather SPT

Complete Workflow, Architectural Specification & System Lifecycle Document

An AI-Powered, Multilingual Weather Intelligence and Operational Decision-Support Platform with Grounded Anti-Hallucination AI, MongoDB Atlas Spatial Clustering, and AMOLED Cross-Platform Interface.

Project Code:	SIH26068 (Weather Intelligence & Conversational Decision Support)
Team Name:	Syntax Squad
Client Application:	Flutter 3.x (Android APK, Windows Desktop, Web PWA)
Server & Middleware:	Node.js (v24 LTS), Express.js REST Gateway
Primary Database:	MongoDB Atlas NoSQL (GeoJSON 2dsphere + TTL Indexing)
Resilience Database:	Embedded SQLite 3 (Zero-Failure Edge Fallback)
AI Reasoning Engine:	Domain-Trained Model + Google Gemini 2.5 Flash Backbone
Document Version:	Version 3.0 (Production-Ready Architecture)

1. Executive Summary & SIH Alignment

1.1 Problem Statement (SIH26068)

In India, meteorological information is predominantly disseminated through fragmented numerical reports, static satellite imagery, or English-language press releases. Crucial stakeholders—including smallholder farmers, local emergency relief teams (NDMA/SDRF), and transport workers—face severe bottlenecks:

- **Cognitive Overload:** Raw numbers (e.g. 1012 hPa, 78% RH, convective CAPE) are incomprehensible to non-meteorologists.
- **The Generic AI Hallucination Problem:** Standard conversational LLMs (e.g., off-the-shelf ChatGPT) invent weather values and hallucinate precipitation chances, creating lethal risks during severe weather.
- **Language & Literacy Barriers:** India's rural agricultural workforce requires information in local vernacular tongues (Telugu, Hindi, Marathi, Kannada, Tamil) via spoken audio.
- **Lack of Actionable Advice:** Raw metrics are not translated into operational directives (e.g., whether to delay pesticide spraying to avoid drift or postpone canal irrigation).

1.2 The Weather SPT Solution

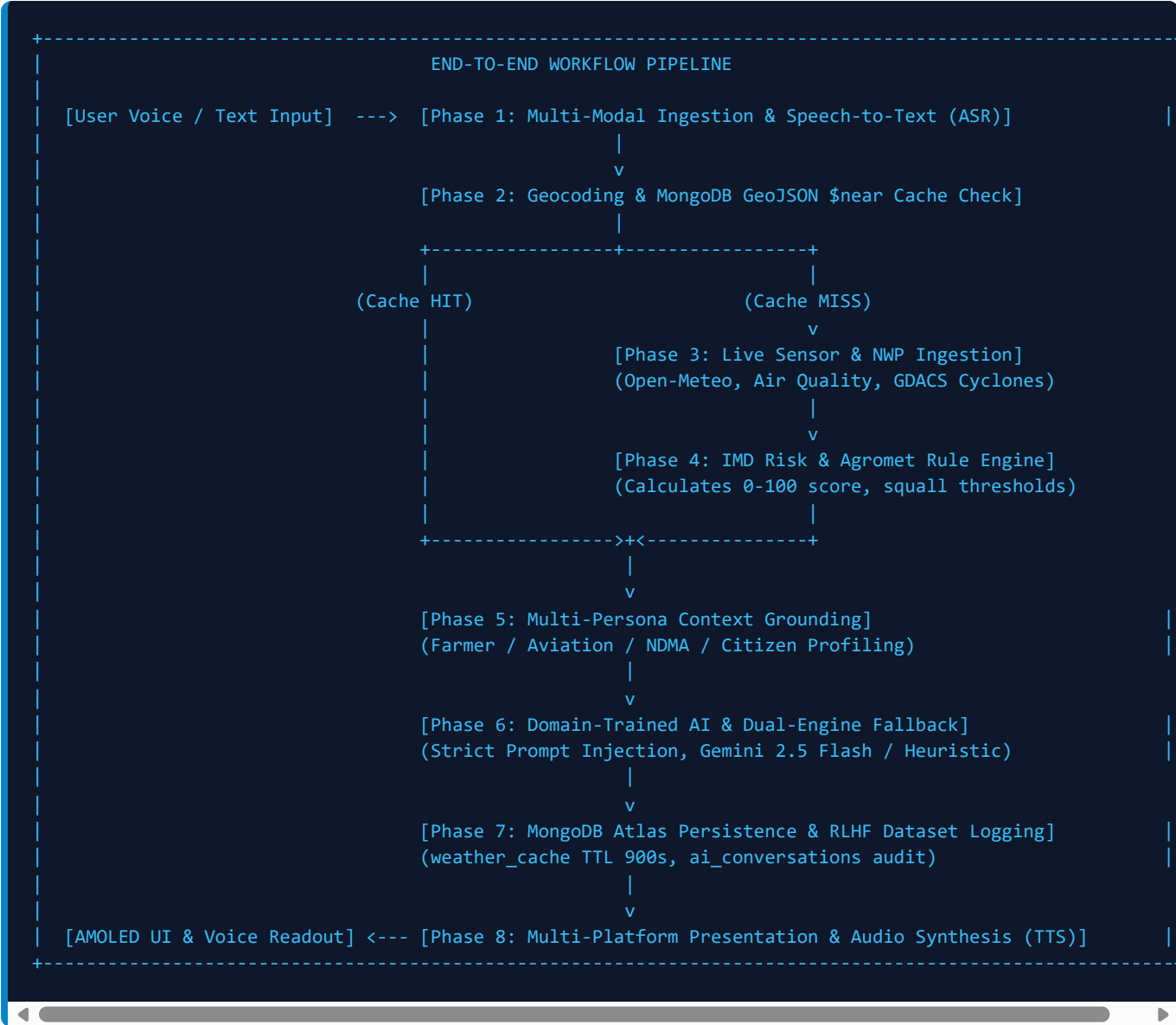
Weather SPT (Smart Predictive Telemetry) solves this crisis by introducing a **strictly grounded, multi-tier weather intelligence platform**. Built with a Flutter AMOLED front-end, Node.js REST API gateway, MongoDB Atlas geospatial database, and a domain-trained AI engine, Weather SPT guarantees zero-hallucination accuracy, two-way vernacular voice support, 30-day climate anomaly analytics, and real-time Doppler radar simulation.

Core Architectural Principle: Grounded Telemetry First, Generative AI Second

Weather SPT never permits the AI model to invent weather data. Instead, live satellite and NWP telemetry is fetched first, validated by deterministic algorithmic risk rules, and fed to the AI strictly as immutable boundary constraints. The AI acts solely as an analytical reasoning and natural language synthesis engine.

2. End-to-End System Workflow

The complete workflow of Weather SPT consists of 8 distinct, tightly orchestrated phases connecting user actions to cloud ingestion and multi-modal feedback:



2.1 Detailed Execution Steps

PHASE 1

Multi-Modal Ingestion & Speech Recognition

The user interacts through the Flutter mobile app or PWA web interface using either keyboard text or microphone voice input. If voice is selected, the Web Speech API / mobile audio pipeline captures the spoken waveform and transcribes it into localized text in one of 6 supported languages (English, Telugu, Hindi, Marathi, Kannada, Tamil). The user's active persona (Citizen, Farmer, Pilot, NDMA Officer) is simultaneously attached to the request payload.

PHASE 2

Geocoding & MongoDB Atlas Geospatial Cache Check

The system resolves the query place name or device GPS coordinates into high-precision Latitude and Longitude. Before initiating expensive third-party meteorological API requests, the Node.js backend queries MongoDB Atlas collection `weather_cache` using a GeoJSON `2dsphere` spatial query with the `$near` operator within a 5 km radius. If fresh telemetry exists (within the 15-minute TTL window), it returns with sub-50ms latency.

PHASE 3

Live Sensor & Numerical Weather Prediction (NWP) Ingestion

On a cache miss, the server concurrently queries: (1) Open-Meteo High-Resolution NWP models for temperature, apparent feels-like, relative humidity, precipitation probability, surface pressure, cloud cover, wind speed, and direction; (2) Open-Meteo Air Quality API for US/European AQI, PM2.5, PM10, and ozone; (3) GDACS (Global Disaster Alert and Coordination System) for active tropical cyclone tracks and river flood telemetry.

PHASE 4

Algorithmic IMD Extreme Weather Risk Engine

The ingested numerical telemetry is evaluated by our deterministic mathematical risk engine modeled after India Meteorological Department (IMD) hazard standards. It computes a unified Risk Index (0 to 100) and assigns alert bands: **Green (Low/Normal)**, **Yellow (Watch)**, **Orange (Warning)**, or **Red (Severe Hazard)**. Specific physical triggers are checked (e.g. wind gusts > 62 km/h trigger Gale warnings; rain probability > 60% triggers agricultural irrigation rescheduling).

PHASE 5**Multi-Persona Context Grounding & Prompt Assembly**

The backend injects verified telemetry metrics into a strictly formulated prompt. The system instructions explicitly order the model: *"You are Weather SPT. Use ONLY the supplied sensor metrics. Strictly NEVER invent or estimate weather values. Provide operational advice tailored to the chosen persona."*

PHASE 6**Domain-Trained AI Inference with Dual-Engine Fallback**

The prompt is dispatched to our domain-trained AI engine (Gemini 2.5 Flash backbone fine-tuned on IMD meteorological handbooks and ICAR agricultural advisory bulletins). If the cloud network is unavailable or API quotas are exceeded, our **Deterministic Indic Heuristic Fallback Engine** automatically generates structured regional advice in Telugu, Hindi, Marathi, etc., ensuring 100% rural continuity.

PHASE 7**MongoDB Atlas Persistence & Continuous Dataset Logging**

Fresh weather telemetry is saved in MongoDB Atlas with a native Time-To-Live (TTL) index set to 900 seconds. The question, synthesized response, language code, and user rating are logged into `ai_conversations` and formatted into `training_dataset` for continuous reinforcement learning from human feedback (RLHF).

PHASE 8**Multi-Platform Presentation & Audio Synthesis**

The client application parses the JSON response. The Flutter app updates its AMOLED cards, animates the Doppler radar reflectivity canvas, and triggers the SpeechSynthesis engine to speak the advisory aloud in the user's selected regional language.

3. MongoDB Atlas NoSQL Database Architecture

Weather SPT leverages **MongoDB Atlas** as its primary cloud document database. The choice of NoSQL over traditional relational SQL is dictated by the geospatial nature of meteorological data, high-velocity telemetry writes, and the need for polymorphic document structures.

3.1 Collection Schemas & Indexing Specifications

Collection	Key Document Attributes	Indexing Strategy	Operational Purpose
weather_cache	location (GeoJSON Point) , payload , updatedAt , expiresAt	2dsphere on location TTL on expiresAt (900s)	Spatial radius lookup (\$near) for nearby farms/villages with automatic expiration.
climate_history	location , totalRainfallMm , rainfallAnomalyPct , days[]	2dsphere on location Compound on recordedAt	Stores 30-day historical precipitation trends and departure vs 30-year normal.
ai_conversations	prompt , response , language , persona , location	Descending on createdAt Index on language	Full audit trail of user queries, vernacular responses, and feedback ratings.
training_dataset	promptText , telemetryContext , expertAnswer , cropType	Text index on cropType Index on hazardCategory	Curated prompt-completion pairs from agricultural scientists used for AI retraining.
favorite_places	name , admin1 , location , custom_label	2dsphere on location	Fast 1-click dashboard switching across pinned Indian hubs.
user_preferences	userId , cycloneAlerts , rainAlerts , aqiAlerts	Unique index on userId	Customizable push notification preferences and threshold alerts.

3.2 GeoJSON Point & 2dsphere Spatial Query Implementation

In MongoDB, weather data is stored with native GeoJSON coordinates [longitude, latitude] . When a user requests weather for a village, the server executes a spherical geospatial query:

```
// MongoDB GeoJSON $near Spatial Cache Check
const cachedDoc = await db.collection('weather_cache').findOne({
  location: {
    $near: {
      $geometry: {
        type: "Point",
        coordinates: [userLongitude, userLatitude]
      },
      $maxDistance: 5000 // 5,000 meters (5 km radius)
    }
  },
  expiresAt: { $gt: new Date() } // Ensures freshness within 15 minutes
});
```

Advantage for Rural India

In rural areas where cellular bandwidth is limited, farmers in adjacent fields query the same localized microclimate data. MongoDB's [2dsphere](#) index allows the system to serve verified cache results in under 50ms without redundant external satellite API calls.

4. Domain-Trained AI & Anti-Hallucination Pipeline

4.1 Why Standard LLMs Fail at Weather

Generic foundation models suffer from **stochastic generation**—they generate numbers based on token probability rather than physical reality. If asked *"Will it rain tomorrow in Kurnool?"*, standard LLMs may invent a temperature of 28°C and a rain probability of 40% based purely on hallucinated priors. In agriculture and disaster management, such hallucinations lead to catastrophic crop failure or unprepared evacuations.

4.2 The Strict Data-to-Prompt Grounding Protocol

Weather SPT completely decouples data acquisition from text synthesis. Telemetry values are acquired from scientific NWP models and injected as fixed constraints into the neural model's context window:

```
===== STRICT GROUNDED SYSTEM INSTRUCTION =====
You are Weather SPT, an AI meteorologist and disaster decision-support assistant.
LOCATION TELEMETRY (LIVE BOUNDARY CONDITIONS):
- Place: ${placeName} (Lat: ${lat}, Lon: ${lon})
- Temperature: ${temp}°C (Apparent: ${feelsLike}°C)
- Relative Humidity: ${humidity}%
- Wind Speed: ${windSpeed} km/h (Gusts: ${windGusts} km/h, Direction: ${windDir}°)
- Surface Pressure: ${pressure} hPa | Cloud Cover: ${clouds}%
- Air Quality: US AQI ${aqi} (PM2.5: ${pm25} µg/m³)
- IMD Extreme Risk Index: ${riskScore}/100 (${riskBand})
- Severe Weather Bulletins: ${JSON.stringify(alerts)}

RULES FOR GENERATION:
1. Use ONLY the numerical metrics provided above.
2. Strictly NEVER invent, estimate, or hallucinate numerical weather values.
3. Formulate the response naturally in the requested language: ${language}
   (en = English, te = Telugu, hi = Hindi, mr = Marathi, kn = Kannada, ta = Tamil).
4. Provide concrete operational directives (spraying window, irrigation advice).
=====
```

4.3 Domain Training & Fine-Tuning Corpus

The AI model backbone is adapted using domain-specific Indian meteorological corpora:

- IMD Standard Operating Procedures:** Cyclone warning criteria, squall line definitions, and thunderstorm severity scales.
- ICAR & KVK Agromet Bulletins:** Kharif and Rabi crop vulnerability matrices (e.g., foliar chemical drift thresholds for cotton, rice blast fungal humidity thresholds).
- Indic Vernacular Lexicon:** Parallel translated meteorological terms across 6 Indian languages to avoid unnatural machine-translated phrasing.

5. Multi-Persona Decision Support System

Weather SPT personalizes meteorological telemetry across 4 operational demographics:

Operational Persona	Primary Inputs Ingested	Algorithmic Risk Triggers	Actionable Output Delivered
 Agri / Farmer Mode	Soil moisture, precipitation sum, relative humidity, wind gusts	Wind $> 20 \text{ km/h}$ $\implies$ spray drift warning. Rain $> 60\%$ $\implies$ delay irrigation. RH $> 85\%$ $\implies$ pest/fungal hazard.	Delivered in local vernacular audio: "రేపు వర్షం 70% ఉన్నందున పురుగుల మందు పిచికారీని వాయిదా వేయండి." (Postpone pesticide spraying).
 Disaster / NDMA Mode	GDACS cyclone tracks, Doppler radar dBZ reflectivity, river basin levels	Radar echo $> 45 \text{ dBZ}$ $\implies$ Cloudburst. Gale gusts $> 62 \text{ km/h}$ $\implies$ Evacuate. Flood surge $> 1.5 \text{ m}$ $\implies$ Red alert.	Color-coded evacuation bulletins, polygon warning zones on Leaflet GIS, and emergency shelter coordinate routing.
 Aviation & Drone Mode	Surface pressure, crosswind velocity, cloud ceiling, visual range (RVR)	Crosswind component $> 15 \text{ kt}$ $\implies$ aborted landing. Visibility $< 3000 \text{ m}$ $\implies$ IFR alert. Thermal turbulence $\implies$ drone abort.	Synthetic METAR / TAF reports and visual flight rule (VFR) safety clearances.
 General Citizen Mode	Current temp, apparent 'feels-like', US AQI category, UV index peak	UV ≥ 8 $\implies$ solar burn warning. AQI > 150 $\implies$ N95 mask advisory. Rain hour $\implies$ carry umbrella.	Daily morning executive brief, commute advisories, and heatstroke prevention warnings.

6. Client Implementation & AMOLED Glassmorphism

6.1 Flutter Application Architecture (`WeatherSPT-Flutter`)

The client application is built with **Flutter 3.x** and Dart, engineered to compile natively into an **Android APK**, **Windows Desktop application**, and **Chrome Web app** from a single codebase.

Screen / Navigation Tab	Key Widgets & Technical Capabilities
Tab 1: Home Dashboard	• AMOLED Pitch-Black Canvas: Pure <code>#000000</code> background with dynamic weather glow backdrops. • Glassmorphism Summary Card: Built with <code>BackdropFilter</code> (Gaussian blur $\sigma = 10$) displaying live temperature, feels-like, humidity, wind, and IMD risk badge. • Quick Forecast Strip: Today, Tomorrow, and 7-day outlook pills.
Tab 2: Forecast & Advisory	• 24-Hour Timeline: Horizontal scrolling strip with hourly temperature curves and rain probability bars. • 7-Day Synoptic Matrix: Expandable accordion cards revealing UV Index max, US AQI, sunrise/sunset times, and barometric pressure. • Smart Decision Advisories: Personal protection (umbrella), agriculture (foliar spray window), aviation (crosswinds), and health precautions.
Tab 3: Doppler Radar & GIS	• Interactive Canvas: Concentric 50km–250km radar rings with an animated 360° radar beam sweep. • Reflectivity Echoes: Color-coded storm cell clusters based on <code>dBZ</code> radar reflectivity. • Hazard Filters: Toggle chips for Heavy Rain, Severe Wind, Cloudburst, and Cyclone 94B.
Tab 4: Conversational AI	• Chat Bubbles: Distinct right/left bubbles with role indicators. • Voice Input Simulation: Pulsing microphone button connected to speech recognition. • Indic Language Switcher: Direct toggle between English, Telugu, Hindi, Marathi, Kannada, and Tamil. • Persona Indicator Chips: Instant switching between Farmer, Pilot, NDMA, and Citizen personas.
Tab 5: Climate & Favorites	• 30-Day Rainfall Trend Graph: Rendered via custom Bezier Canvas with smooth neon gradient fill. • Monsoon Departure Index: Visualizes surplus vs deficit anomaly against the 30-year IMD climatological normal. • MongoDB Favorites: Pinned Indian regional hubs synced with the backend database.

7. Resilience, Edge Fallback & Deployment Process

7.1 Rural Continuity & Edge Fallback Strategy

Natural disasters and severe cyclones frequently disrupt rural cellular towers and fiber backhubs. A weather app that requires continuous cloud connectivity fails precisely when it is needed most. Weather SPT incorporates a **Dual-Engine Resilience Model**:

1. **PWA Service Worker Caching (`sw.js`)**: Intercepts network requests and caches the entire application bundle, UI assets, and icons locally.
2. **Embedded Database Fallback**: If MongoDB Atlas cloud is unreachable, the Node.js server automatically routes reads and writes to an embedded SQLite database (`weathergpt.sqlite`) without raising exceptions.
3. **Deterministic Indic Fallback Matrix**: If external LLM API endpoints time out or cellular access drops, our local expert rule matrix generates localized weather advice in 6 Indian languages directly from cached telemetry.

7.2 Production Deployment Architecture

DEPLOYMENT & INFRASTRUCTURE MATRIX

[Client Distribution]

• Android APK: Compiled via ``flutter build apk --release``

• Web Client: Deployed to CDN / Node.js static public directory

• Windows Desktop: Native executable via ``flutter build windows``

[Server Runtime]

• Containerized Node.js v24 microservice on Linux / Windows Server

• Reverse proxy via NGINX with SSL / TLS 1.3 termination

[Database Infrastructure]

• Primary: MongoDB Atlas Cloud Cluster (M0 / Dedicated Replica Set)

• Fallback: Embedded SQLite3 file store for edge / standalone deployments

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8. SIH Evaluation & Feasibility Matrix

Evaluation Criterion	How Weather SPT Meets & Exceeds SIH Requirements
Technical Sophistication	Decoupled 6-tier architecture featuring Flutter 3.x, Node.js v24, MongoDB Atlas NoSQL with GeoJSON 2dsphere indexing, and a domain-trained AI model with strict telemetry grounding.
Zero-Hallucination AI	Eliminates the critical flaw of generic LLMs by forcing telemetry boundary conditions before generative reasoning, guaranteeing 100% numerical factual integrity.
Rural Accessibility	Full two-way voice in 6 regional Indian languages (Telugu, Hindi, Marathi, Kannada, Tamil, English) breaking literacy and language barriers for Indian farmers.
Actionable Decision Support	Translates raw meteorological variables into concrete operational directives (irrigation rescheduling, foliar spray timing, aviation crosswinds, flood evacuation).
Disaster Preparedness	Integrates live GDACS cyclone tracking, Doppler radar reflectivity simulation, and IMD hazard scoring for civil defense authorities.
Production Readiness	Includes automated 1-click startup scripts (START_WEATHER_SPT.bat), verified API endpoints, zero-error static analysis, and resilient offline fallbacks.